PADHAKU
Science · Class 10 Science



CHAPTER 3

Metals and Non-metals

NCERT Exercise Solutions

Think about it · Introduction	3 Q
Activities · 3.1.1 Metals	9 Q
Activities · 3.1.2 Non-metals	2 Q
Questions · 3.1 Physical Properties	2 Q
Activities · 3.2 Chemical Properties of Metals	5 Q
Questions · 3.2 Chemical Properties of Metals	5 Q
In-text · 3.3 How do Metals and Non-metals React?	2 Q
Questions · 3.3 How do Metals and Non-metals React?	2 Q
In-text · 3.4 Occurrence of Metals	1 Q
Questions · 3.4 Occurrence of Metals	3 Q
Activity · 3.5 Corrosion	2 Q
Questions · 3.5 Corrosion	3 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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🤔 Think about it · Introduction

? Think of some uses of metals and non-metals in your daily life.

💡 SAMPLE ANSWER

- Metals: iron for tools and gates, copper for electric wires, aluminium for cooking vessels and foil, gold and silver for jewellery.
- Non-metals: oxygen for breathing, carbon (coal) as fuel, chlorine to clean water, sulphur in medicines and crackers.

? What properties did you think of while categorising elements as metals or non-metals?

💡 SAMPLE ANSWER

Lustre (shine), hardness, malleability, ductility, conduction of heat and electricity, sonority (ringing sound) and the state at room temperature.

? How are these properties related to the uses of these elements?

💡 SAMPLE ANSWER

- Copper and aluminium conduct electricity, so they are used for wires.
- Aluminium and copper conduct heat, so they are used for cooking vessels.
- Gold and silver are lustrous and malleable, so they are used for jewellery.
- Metals are sonorous, so bells are made of metal.

⭐ Activities · 3.1.1 Metals

Activity 3.1 **Note the appearance of iron, copper, aluminium and magnesium before and after rubbing them with sand paper.**

✔ ANSWER

Observation: Before rubbing, the surfaces look dull. After rubbing, they look shiny.

Conclusion: Metals in their pure state have a shining surface. This is called **metallic lustre**.

Activity 3.2 **Try to cut iron, copper, aluminium, magnesium and sodium with a sharp knife. What do you observe?**

✔ ANSWER

Observation: Iron, copper, aluminium and magnesium cannot be cut with a knife. Sodium is cut easily.

Conclusion: Metals are generally hard, but hardness varies. Sodium (an alkali metal) is very soft.

Activity 3.3 **Strike iron, zinc, lead and copper with a hammer on an iron block. What do you observe?**

✔ ANSWER

Observation: The metals become flat and thin; they do not break.

Conclusion: Metals can be beaten into thin sheets. This property is called **malleability**.

Activity 3.4 **List the metals whose wires you have seen in daily life.**

✔ ANSWER

Copper, aluminium, iron (steel), silver and gold (wires used in jewellery).

Conclusion: Metals can be drawn into thin wires. This property is called **ductility**.

? **Can you name some metals that are used for making cooking vessels? Why are these metals used?**

✔ ANSWER

Aluminium, copper, iron and steel (an alloy of iron).

They are good conductors of heat and have high melting points, so they heat food quickly and do not melt on the flame.

Activity 3.5 **Heat an aluminium or copper wire (with a pin fixed by wax at its free end) near the clamp. What do you observe? Does the wire melt?**

✔ ANSWER

Observation: After some time the wax melts and the pin falls. The wire does not melt.

Conclusion: Metals are good conductors of heat and have high melting points.

Activity 3.6 **Place the metal between terminals A and B of the circuit. Does the bulb glow? What does this indicate?**

✔ ANSWER

Observation: Yes, the bulb glows.

Conclusion: Metals are good conductors of electricity.

? **Why are electric wires coated with PVC or a rubber-like material?**

✔ ANSWER

PVC and rubber are bad conductors (insulators) of electricity. The coating prevents electric shock and short circuits when wires touch us or each other.

? **Can you now say why school bells are made of metals?**

✔ ANSWER

Metals are **sonorous**: they produce a ringing sound when struck on a hard surface.

⭐ Activities · 3.1.2 Non-metals

Activity 3.7 **Carry out Activities 3.1 to 3.4 and 3.6 with carbon, sulphur and iodine and record the observations in Table 3.1.**

💡 SAMPLE ANSWER

Element	Symbol	Surface	Hardness	Malleability	Ductility	Conducts electricity	Sonority
Iron	Fe	Lustrous	Hard	Yes	Yes	Yes	Yes
Copper	Cu	Lustrous	Hard	Yes	Yes	Yes	Yes
Sodium	Na	Lustrous (when cut)	Soft	Yes	Yes	Yes	—
Carbon (coal)	C	Dull	Brittle	No	No	No (graphite: yes)	No
Sulphur	S	Dull	Brittle	No	No	No	No
Iodine	I	Lustrous	Brittle	No	No	No	No

Conclusion: Non-metals are generally dull, brittle, not malleable or ductile and are bad conductors. But there are exceptions (iodine is lustrous, graphite conducts), so elements cannot be grouped by physical properties alone.

Activity 3.8 **Burn magnesium ribbon and sulphur powder, dissolve the products in water and test with litmus. Are the products acidic or basic? Write the equations.**

✔ ANSWER

Magnesium: burns with a dazzling white flame and forms white ash (magnesium oxide). Its solution turns red litmus **blue**, so the product is **basic**.

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(aq)$

Sulphur: burns to give sulphur dioxide fumes. Its solution turns blue litmus **red**, so the product is **acidic**.

$S(s) + O_2(g) \rightarrow SO_2(g)$

$SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$

Conclusion: Metals form basic oxides; non-metals form acidic oxides.

❓ Questions · 3.1 Physical Properties

Q 1 **Give an example of a metal which (i) is a liquid at room temperature (ii) can be easily cut with a knife (iii) is the best conductor of heat (iv) is a poor conductor of heat.**

✔ ANSWER

(i) **Mercury**

(ii) **Sodium** (also lithium, potassium)

(iii) **Silver** (copper is next)

(iv) **Lead** (also mercury)

Q 2 **Explain the meanings of malleable and ductile.**

✔ ANSWER

- **Malleable:** a metal that can be beaten into thin sheets (e.g. gold, silver, aluminium).
- **Ductile:** a metal that can be drawn into thin wires (e.g. gold, copper, aluminium).

⭐ Activities · 3.2 Chemical Properties of Metals

Activity 3.9 **Burn samples of aluminium, copper, iron, lead, magnesium, zinc and sodium over a flame. Which burn easily? Flame colour? Surface after burning? Order of reactivity towards oxygen? Are the products soluble in water?**

✔ ANSWER

- **Burn easily:** sodium and magnesium.
- **Flame colour:** sodium – golden yellow; magnesium – dazzling white.
- **Surface:** becomes dull, covered with a layer of oxide (copper gets a black layer of copper(II) oxide).
- **Order:** Na > Mg > (Al, Zn, Fe, Pb, Cu – these hardly burn, so their order cannot be decided by this test).
- **Solubility:** sodium oxide dissolves to form an alkali (NaOH); magnesium oxide dissolves slightly; most other metal oxides are insoluble.

Activity 3.10 **Put the metals in cold water, then hot water, then steam. Which react? Which produce fire or float? Arrange them in decreasing order of reactivity with water.**

✔ ANSWER

- **Cold water:** sodium reacts violently; the hydrogen formed catches fire.
- **Hot water:** magnesium reacts and starts floating (hydrogen bubbles stick to it).
- **Steam:** aluminium, zinc and iron react to form oxide and hydrogen.
- **No reaction even with steam:** lead and copper.
- **Decreasing order: Na > Mg > Al > Zn > Fe** (Pb and Cu do not react).

Activity 3.11 **Put the metals (except Na and K) in dilute HCl. Which react vigorously? Which gives the highest temperature? Arrange in decreasing order of reactivity. Write equations for Mg, Al, Zn and Fe.**

✔ ANSWER

- **Most vigorous / highest temperature:** magnesium (bubbles form fastest).
- **Order: Mg > Al > Zn > Fe**; lead reacts very slowly; copper does not react (no bubbles, no change in temperature).

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$

Activity 3.12 **Put a copper wire in iron sulphate solution and an iron nail in copper sulphate solution. Observe after 20 minutes.**

✔ ANSWER

- **Reaction occurs** in the test tube with the **iron nail in copper sulphate**.
- **Basis:** the blue colour of the solution fades (becomes pale green) and a reddish-brown coating of copper forms on the nail. No change is seen with the copper wire.
- **Correlation:** iron is more reactive than copper, as also seen in Activities 3.9–3.11 (copper did not react with water or dilute acid; iron did).

$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

- **Type:** displacement reaction.

? **Which metal, copper or iron, is more reactive according to your observations in Activity 3.12?**

✔ ANSWER

Iron is more reactive, because it displaces copper from copper sulphate solution.

❓ Questions · 3.2 Chemical Properties of Metals

Q 1 **Why is sodium kept immersed in kerosene oil?**

✔ ANSWER

Sodium is very reactive. It reacts so vigorously with oxygen and moisture of air that it catches fire if kept in the open.

Kerosene does not react with sodium and keeps air away, so it is stored in kerosene to prevent accidental fire.

Q 2 **Write equations for the reactions of (i) iron with steam (ii) calcium and potassium with water.**

✔ ANSWER

(i) $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

(ii) $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$

$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$ + heat energy

Q 3 **Four metals A, B, C and D were added to these solutions:**

Metal	Iron(II) sulphate	Copper(II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement	—	—
B	Displacement	—	No reaction	—
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

(i) Which is the most reactive metal? (ii) What would you observe if B is added to copper(II) sulphate solution? (iii) Arrange A, B, C and D in decreasing order of reactivity.

✔ ANSWER

Reasoning: B displaces iron but not zinc (between Zn and Fe). A displaces copper but not iron (between Fe and Cu). C displaces only silver (between Cu and Ag). D displaces none (less reactive than Ag).

(i) **B** is the most reactive.

(ii) B is more reactive than iron, so also more reactive than copper. It will **displace copper**: the blue colour of the solution fades and a reddish-brown layer of copper is deposited on B.

(iii) **B > A > C > D**

Q 4 **Which gas is produced when dilute hydrochloric acid is added to a reactive metal? Write the chemical reaction when iron reacts with dilute H_2SO_4.**

✔ ANSWER

Hydrogen gas is produced.

$Fe(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2(g)$

Q 5 **What would you observe when zinc is added to a solution of iron(II) sulphate? Write the chemical reaction.**

✔ ANSWER

Zinc is more reactive than iron, so it displaces iron. The pale green solution becomes colourless and a dark grey/black deposit of iron forms on the zinc.

$Zn(s) + FeSO_4(aq) \rightarrow ZnSO_4(aq) + Fe(s)$

⭐ In-text · 3.3 How do Metals and Non-metals React?

? **Can you name the cation and anion present in $MgCl_2$?**

✔ ANSWER

Cation: **magnesium ion, Mg^{2+}**. Anion: **chloride ion, Cl^-** (two Cl^- for each Mg^{2+}).

Activity 3.13 **Take sodium chloride, potassium iodide, barium chloride etc. Note the physical state; heat on a spatula; dissolve in water, petrol and kerosene; test the solution in a circuit. What is your inference?**

✔ ANSWER

- **State:** solid, hard crystals.
- **Heating:** they do not melt easily (very high melting points). They give a colour to the flame, e.g. sodium chloride – golden yellow.
- **Solubility:** soluble in water; insoluble in petrol and kerosene.
- **Circuit:** the bulb glows with their solutions in water, so the solutions conduct electricity.
- **Inference:** these are **ionic compounds** made of positive and negative ions.

❓ Questions · 3.3 How do Metals and Non-metals React?

Q 1 **(i) Write the electron-dot structures for sodium, oxygen and magnesium. (ii) Show the formation of Na_2O and MgO by the transfer of electrons. (iii) What are the ions present in these compounds?**

✔ ANSWER

(i) Only valence electrons are shown as dots:

- Sodium (2,8,1): **Na·** (1 dot)
- Oxygen (2,6): **:Ö:** (6 dots around O)
- Magnesium (2,8,2): **·Mg·** (2 dots)

(ii) **Na_2O:** each of two Na atoms gives 1 electron to one O atom.

$2Na \rightarrow 2Na^+ + 2e^-$ (2,8,1 → 2,8)

$O + 2e^- \rightarrow O^{2-}$ (2,6 → 2,8)

$2Na^+ + O^{2-} \rightarrow Na_2O$, i.e. $Na^+ [:Ö:]^{2-} Na^+$

MgO: one Mg atom gives 2 electrons to one O atom.

$Mg \rightarrow Mg^{2+} + 2e^-$ (2,8,2 → 2,8)

$O + 2e^- \rightarrow O^{2-}$ (2,6 → 2,8)

$Mg^{2+} + O^{2-} \rightarrow MgO$, i.e. $Mg^{2+} [:Ö:]^{2-}$

(iii) Na_2O: **Na^+ and O^{2-}** ions. MgO: **Mg^{2+} and O^{2-}** ions.

Q 2 **Why do ionic compounds have high melting points?**

✔ ANSWER

Ionic compounds have strong electrostatic forces of attraction between oppositely charged ions. A large amount of energy is needed to break this strong inter-ionic attraction, so their melting points are high.

⭐ In-text · 3.4 Occurrence of Metals

? **$3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + Heat$**

Can you identify the substances that are getting oxidised and reduced?

✔ ANSWER

- **Oxidised:** aluminium (Al) – it gains oxygen to form Al_2O_3.
- **Reduced:** manganese dioxide (MnO_2) – it loses oxygen to form Mn.

❓ Questions · 3.4 Occurrence of Metals

Q 1 **Define the following terms. (i) Mineral (ii) Ore (iii) Gangue**

✔ ANSWER

(i) **Mineral:** an element or compound that occurs naturally in the earth's crust.

(ii) **Ore:** a mineral that has a high percentage of a metal, from which the metal can be extracted profitably.

(iii) **Gangue:** impurities such as soil and sand present in the ore mined from the earth.

Q 2 **Name two metals which are found in nature in the free state.**

✔ ANSWER

Gold and silver (also platinum, copper).

Q 3 **What chemical process is used for obtaining a metal from its oxide?**

✔ ANSWER

Reduction – e.g. heating the oxide with carbon: ZnO(s) + C(s) → Zn(s) + CO(g). Highly reactive metals are obtained by electrolytic reduction; very reactive metals can also be used to reduce oxides (e.g. thermit reaction).

🧪 Activity · 3.5 Corrosion

Activity 3.14 **Iron nails are kept in test tube A (water + air), B (boiled distilled water with oil layer) and C (dry air with anhydrous calcium chloride). What does this tell us about the conditions under which iron rusts?**

✔ ANSWER

Observation: Nails rust only in test tube A; they do not rust in B (no air) or C (no moisture).

Conclusion: Iron rusts only when **both air (oxygen) and water (moisture)** are present.

? **The galvanised article is protected against rusting even if the zinc coating is broken. Can you reason this out?**

✔ ANSWER

Zinc is more reactive than iron. If the coating breaks, zinc still reacts with oxygen and moisture first (in place of iron), so the iron is protected.

❓ Questions · 3.5 Corrosion

Q 1 **Metallic oxides of zinc, magnesium and copper were heated with the following metals:**

Oxide \ Metal	Zinc	Magnesium	Copper
Zinc oxide			
Magnesium oxide			

Copper oxide		

In which cases will you find displacement reactions taking place?

✔ ANSWER

A metal displaces another metal from its oxide only if it is more reactive (Mg > Zn > Cu).

Oxide \ Metal	Zinc	Magnesium	Copper
Zinc oxide	No	Yes	No
Magnesium oxide	No	No	No
Copper oxide	Yes	Yes	No

Displacement takes place in: **zinc + copper oxide, magnesium + zinc oxide, magnesium + copper oxide**.

Zn + CuO → ZnO + Cu; Mg + ZnO → MgO + Zn; Mg + CuO → MgO + Cu

Q 2 **Which metals do not corrode easily?**

✔ ANSWER

Less reactive metals at the bottom of the activity series, such as **gold and platinum**, do not corrode easily. (Silver only tarnishes slowly.)

Q 3 **What are alloys?**

✔ ANSWER

An **alloy** is a homogeneous mixture of two or more metals, or of a metal and a non-metal. Examples: brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn), stainless steel (Fe + Ni + Cr).

📝 Exercises

1 **Which of the following pairs will give displacement reactions? (a) NaCl solution and copper metal (b) $MgCl_2$ solution and aluminium metal (c) $FeSO_4$ solution and silver metal (d) $AgNO_3$ solution and copper metal.**

✔ ANSWER

(d) $AgNO_3$ solution and copper metal. Copper is more reactive than silver, so it displaces silver.

$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$

2 **Which of the following methods is suitable for preventing an iron frying pan from rusting? (a) Applying grease (b) Applying paint (c) Applying a coating of zinc (d) All of the above.**

✔ ANSWER

(c) Applying a coating of zinc. A frying pan is heated, so grease and paint would burn off; a zinc coating (galvanising) stays and protects the iron.

3 **An element reacts with oxygen to give a compound with a high melting point. This compound is also soluble in water. The element is likely to be (a) calcium (b) carbon (c) silicon (d) iron.**

✔ ANSWER

(a) calcium. Calcium oxide (CaO) is an ionic compound with a high melting point (2850 K) and dissolves in water to form calcium hydroxide.

4 **Food cans are coated with tin and not with zinc because (a) zinc is costlier than tin (b) zinc has a higher melting point than tin (c) zinc is more reactive than tin (d) zinc is less reactive than tin.**

✔ ANSWER

(c) zinc is more reactive than tin. Zinc could react with acids in the food; tin is less reactive and safe.

5 **You are given a hammer, a battery, a bulb, wires and a switch. (a) How could you use them to distinguish between samples of metals and non-metals? (b) Assess the usefulness of these tests.**

✔ ANSWER

(a)

- **Hammer test (malleability):** beat each sample. Metals become flat sheets; non-metals break into pieces (brittle).
- **Conductivity test:** connect the sample between two wires in a circuit with the battery, bulb and switch. If the bulb glows, the sample is a metal; if not, it is a non-metal.

(b) These tests are useful and easy, but not fully reliable because of exceptions:

- Graphite (a non-metal) conducts electricity.
- Sodium and potassium are very soft and mercury is a liquid, so the hammer test does not work well for them.

So chemical properties must also be checked.

6 **What are amphoteric oxides? Give two examples of amphoteric oxides.**

✔ ANSWER

Metal oxides that react with both acids and bases to produce salt and water are called **amphoteric oxides**.

Examples: **aluminium oxide (Al_2O_3)** and **zinc oxide (ZnO)**.

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$

$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$

7 **Name two metals which will displace hydrogen from dilute acids, and two metals which will not.**

✔ ANSWER

- **Displace hydrogen:** magnesium and zinc (also iron, aluminium) – they are above hydrogen in the activity series.
- **Do not displace hydrogen:** copper and silver (also gold) – they are below hydrogen.

8 **In the electrolytic refining of a metal M, what would you take as the anode, the cathode and the electrolyte?**

✔ ANSWER

- **Anode:** a block of impure metal M.
- **Cathode:** a thin strip of pure metal M.
- **Electrolyte:** a solution of a salt of metal M (e.g. for copper, acidified copper sulphate solution).

9 **Pratyush heated sulphur powder and collected the gas in an inverted test tube. (a) What will be the action of the gas on (i) dry litmus paper (ii) moist litmus paper? (b) Write a balanced chemical equation for the reaction.**

✔ ANSWER

The gas is **sulphur dioxide (SO_2)**.

(a) (i) **Dry litmus paper:** no change (SO_2 needs water to form an acid).

(ii) **Moist blue litmus paper turns red**, because SO_2 dissolves in water to form sulphurous acid: $SO_2 + H_2O \rightarrow H_2SO_3$.

(b) $S(s) + O_2(g) \rightarrow SO_2(g)$

10 **State two ways to prevent the rusting of iron.**

✔ ANSWER

- **Painting, oiling or greasing** – keeps air and moisture away.
- **Galvanising** – coating iron with a thin layer of zinc.

(Others: chrome plating, making alloys such as stainless steel.)

11 **What type of oxides are formed when non-metals combine with oxygen?**

✔ ANSWER

Non-metals form **acidic oxides** (e.g. SO_2, CO_2) or **neutral oxides** (e.g. CO, H_2O).

12 (a) **Give reasons: Platinum, gold and silver are used to make jewellery.**

✔ ANSWER

They are very lustrous, malleable and ductile, and they are very less reactive, so they do not corrode or lose their shine easily.

12 (b) **Give reasons: Sodium, potassium and lithium are stored under oil.**

✔ ANSWER

They are highly reactive and react vigorously with oxygen and moisture in air (sodium and potassium even catch fire). Oil keeps air and water away.

12 (c) **Give reasons: Aluminium is a highly reactive metal, yet it is used to make utensils for cooking.**

✔ ANSWER

Aluminium gets covered with a thin protective layer of aluminium oxide, which stops further reaction. It is also light, a good conductor of heat and has a high melting point.

12 (d) **Give reasons: Carbonate and sulphide ores are usually converted into oxides during the process of extraction.**

✔ ANSWER

It is easier to obtain a metal from its oxide (by reduction with carbon) than from its sulphide or carbonate. So sulphide ores are roasted and carbonate ores are calcined to get oxides.

$2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$ (roasting)

$ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)$ (calcination)

13 **You must have seen tarnished copper vessels being cleaned with lemon or tamarind juice. Explain why these sour substances are effective in cleaning the vessels.**

✔ ANSWER

Copper reacts with moist carbon dioxide of air to form a green coat of **basic copper carbonate**, which is basic in nature.

Lemon and tamarind contain **acids**. The acid reacts with (neutralises) and dissolves the basic copper carbonate layer, so the shiny copper surface appears again.

14 **Differentiate between metal and non-metal on the basis of their chemical properties.**

✔ ANSWER

Metals	Non-metals
Form basic oxides (some, like Al_2O_3 and ZnO, are amphoteric)	Form acidic or neutral oxides
Lose electrons to form positive ions (cations), e.g. Na^+	Gain electrons to form negative ions (anions), e.g. Cl^-
Many react with water to give metal oxide/hydroxide and hydrogen	Generally do not react with water
Metals above hydrogen displace hydrogen from dilute acids	Do not displace hydrogen from dilute acids
More reactive metals displace less reactive metals from their salt solutions	React with hydrogen to form hydrides

15 **A man posing as a goldsmith dipped gold bangles in a particular solution. They sparkled like new but their weight was reduced drastically. Find out the nature of the solution he had used.**

✔ ANSWER

He used **aqua regia**, a freshly prepared mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio **3 : 1**.

Aqua regia dissolves gold. It dissolved the outer layer of the bangles, so the fresh gold below shone, but the weight decreased.

16 **Give reasons why copper is used to make hot water tanks and not steel (an alloy of iron).**

✔ ANSWER

- Copper does not react with water, even hot water or steam, so the tank does not corrode.
- Iron (in steel) reacts with steam and rusts in the presence of water and air, so a steel tank would corrode.
- Copper is also a very good conductor of heat, so water heats quickly.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**